SILAS KWABLA GAH

PhD Candidate in Computer Vision · 3D Geometric Vision, Generative Modeling & Calibration

Department of Computer Science, University of Ghana, Legon, Accra, Ghana

gahsilas@gmail.com | +233 558 075 023 | silsgah.github.io | GitHub | LinkedIn

Research Profile

Final-year PhD candidate in Computer Vision and Machine Learning at the University of Ghana (thesis defence scheduled for October 2026). Research specializes in **3D geometric vision**, **inference-time foundation-model composition**, **distribution-preserving semantic retention**, and **probabilistic calibration under extreme data scarcity**. Experienced in processing dense 3D point clouds, volumetric representations, multi-view camera projective geometry (K, T), and self-supervised process rewards. Proven track record in solving low-data perception problems without parameter degradation ($\Delta\theta = 0$). Directly aligned with the **Generative AI & 3D Defect Synthesis** initiative at KAUST: developing physics-aware generative models (Diffusion/NeRF/implicit fields), synthesizing complex 3D defect morphologies (cracks, voids, inclusions), and calibrating volumetric anomaly detection pipelines.

Education

PhD in Computer Vision and Machine Learning — **Expected defence: October 2026**
University of Ghana, Legon — *Supervisor: Prof. Ebenezer Owusu*

Thesis: *Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations*

Investigates 3D geometric point-cloud segmentation under few-shot supervision, non-parametric evidence coordination, avoiding premature decision collapse, and reliability-calibrated inference.

MPhil in Computer Science — **2016**
University of Ghana, Legon — *Accra, Ghana*

BSc in Information and Communications Technology — **Second Class Honours (Upper Division) · 2012**
Ghana Institute of Management and Public Administration (GIMPA) — *Accra, Ghana*

HND in Electrical & Electronic Engineering — **2009**
Accra Technical University — *Accra, Ghana*

Doctoral Research Highlights & Systems Rigor

Dense 3D Point-Cloud Perception & Geometric Consistency (ACCV 2026, WACV 2027, ICLR 2027)

- Developed training-free open-vocabulary 3D segmentation architectures reconciling 3D spatial representations with multi-view 2D promptable segmenters (SAM 2.1) via camera-pose ray casting and geometric consensus.
- Evaluated on massive 3D benchmarks (**ScanNet200**, **ScanNet++**, 312 scenes, ∼50M points), achieving state-of-the-art harmonic mean (34.87 novel mIoU, +6.40 points improvement) with paired bootstrap 95% confidence intervals [5.24, 7.64].
- Discovered and formalized the mechanism of **premature semantic collapse** under hard argmax selection; proved that retaining compact top-k candidate distributions preserves subtle, fine-grained class alternatives that standard pipelines discard (formalized in an **ICLR 2027** submission).
- Engineered high-throughput, memory-bounded evaluation pipelines with float16 caching and chunked processing to evaluate dense 3D spatial distributions on constrained GPU clusters.

Reliability Calibration & Uncertainty Quantification (RCASA) (Target: IEEE TPAMI In Prep)

- Formulated **RCASA**, a principled probabilistic framework that decouples classifier confidence from contextual reliability via Platt scaling and Bernoulli uncertainty estimation.
- Proved class-wise bounded updates and probability simplex preservation, providing rigorous trust guarantees for detecting rare anomalies and out-of-distribution patterns without false positives.

Multi-View Physical Consistency as Intrinsic Self-Supervision (SACAIR 2026 Submission)

- Formulated physical consistency across independent viewpoints as an intrinsic, label-free process reward for multimodal reasoning models (**Qwen2.5-VL-7B**) using online **GRPO/VERL** on **vLLM**.
- Mitigated reward-hacking failure modes through counterfactual stress-testing, eliminating spatial area bias ($\rho > 0.72 \rightarrow 0.334$) without human-labeled reward annotations.

Selected Publications and Manuscripts

- **Beyond Argmax: A Mechanistic Study of Semantic Retention in Frozen Foundation-Model Composition for Generalized Few-Shot 3D Segmentation.**
  International Conference on Learning Representations (ICLR 2027 Conference Submission; Preprint Available). First Author.
- **Training-Free Open-Vocabulary 3D Point-Cloud Segmentation on the Generalized Few-Shot Benchmark.**
  Asian Conference on Computer Vision (ACCV 2026, Paper #482; Preprint Available). First Author.
- **Training-Free Generalised Few-Shot Segmentation through Open-Vocabulary Semantic Arbitration.**
  IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027, Paper #167). First Author.
- **Cross-View Agreement as a Label-Free Process Reward for Multimodal Reasoning.**
  Southern African Conference for AI Research (SACAIR 2026 Submission). First Author.
- **Transfer Learning-Based Encoder–Decoder Model for Skin Lesion Segmentation.**
  Communications in Computer and Information Science (CCIS), Vol. 1841, pp. 117–128, Springer, 2023. Co-Author.
- **Reliability-Calibrated Adaptive Semantic Arbitration of Frozen Foundation Model Representations.**
  Manuscript in preparation (targeted for IEEE TPAMI). First Author.

Technical Skills

Programming	Python, C++, Java
Deep Learning & RL	PyTorch, Transformers, vLLM, Diffusion models, LoRA/QLoRA, SFT, GRPO, VERL
3D & Volumetric Vision	3D point clouds, volumetric grids, camera projection (K, T), ray casting, 2D–3D lifting, multi-view consensus, ScanNet200/ScanNet++
Generative Modeling	Diffusion-based generation, NeRF fundamentals, Top-k distribution modeling, anomaly detection
Statistical Calibration	Platt scaling, Expected Calibration Error (ECE), Brier score, paired bootstrap CIs, uncertainty estimation
Systems & GPU	CUDA fundamentals, Triton kernel benchmarking, HPC cluster environments, streaming evaluation

Teaching, Professional Experience & Academic Service

Part-time Lecturer — Accra Technical University (2022 – Present)

- Lecturing in algorithms, machine learning, and data structures; previously taught at GIMPA.

Founder & Lead Architect — EED Soft Consult (2018 – Present)

- Architected and deployed production enterprise and distributed software platforms in Ghana.

Academic Service — Reviewer, SACAIR 2026 Program Committee

References

Prof. Ebenezer Owusu (PhD Supervisor)
Associate Professor, Department of Computer Science
University of Ghana, Legon
Email: eowusu@ug.edu.gh

Prof. Jamal-Deen Abdulai (Head of Department)
Associate Professor, Department of Computer Science
University of Ghana, Legon
Email: jabdulai@ug.edu.gh